COLLEGE OF ENGINEERING TRIVANDRUM

Department of Computer Applications



20MCA243

MOBILE APPLICATION DEVELOPMENT LAB

Name: **DEEPAK H**

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Certified Bonafide Record of Work Done By DEEPAK H

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External Examiner

Faculty In-Charge

Prof. Najna S

Contents

App 1 : Login Page 1
App2 : Simple Calculator 2
App 3 : Intent 3
App 4 : Activity Life Cycle 4
App 5 : Check Box and Radio Button 5
App 6 : Implicit Intent 6
App 7 : List View 7
App 8 : Scroll View 8
App 9 : Spinner 9
App 10 : Shared Preferences 10

Date: 08/07/25

APP 1: Login Page

Aim: Design a Login page with username and password using Linear Layout and Toast Valid and Invalid credentials.

Program Code:

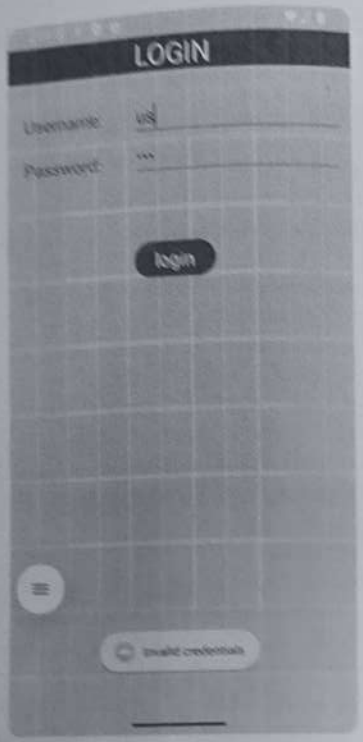
MainActivity.java

```
package com.example.myapplication;
import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
   Button b;
   EditText u,p;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      b=(Button) findViewById(R.id.b);
      u=(EditText) findViewById(R.id.user);
      p=(EditText) findViewById(R.id.pass);
      b.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View view) {
            String user=u.getText().toString();
            String pass=p.getText().toString();
            if (user.equals("usr")&& pass.equals("123"))
               Toast.makeText(getApplicationContext(),"logged in
successfully",Toast.LENGTH_SHORT).show();
            else
               Toast.makeText(getApplicationContext(),"Invalid
credentials",Toast.LENGTH_SHORT).show();
         }})
   }
}
```





Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity"
android:background="@drawable/ic_launcher_background">
   <RelativeLayout
      android:layout_width="match_parent"
      android:layout_height="match_parent">

      <TextView
         android:id="@+id/T"
         android:layout_width="1200dp"
         android:layout_height="wrap_content"
         android:layout_alignParentEnd="true"
         android:layout_marginTop="25dp"
         android:layout_marginEnd="0dp"
         android:background="#B23B1E40"
         android:text="LOGIN"
         android:textAlignment="center"
         android:textColor="#ffffff"
         android:textSize="30dp" />

      <TextView
         android:id="@+id/u"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:text="Username:"
         android:layout_marginTop="100dp"
         android:layout_marginLeft="20dp"
         android:textSize="20dp"
         />

<TextView
         android:id="@+id/textView3"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:text="Password:"
         android:layout_marginTop="150dp"
         android:layout_marginLeft="20dp"
         android:textSize="20dp"
         />
<Button
         android:id="@+id/b"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:layout_marginLeft="150dp"
         android:layout_marginTop="250dp"
android:backgroundTint="#DF803BBC"
         android:text="login"
         android:textSize="20dp" />

      <EditText
         android:id="@+id/user"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:layout_marginTop="90dp"
         android:layout_marginLeft="150dp"
         android:textSize="20dp"
         android:ems="10"
         android:inputType="text"
         android:text="" />

      <EditText
         android:id="@+id/pass"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:layout_marginTop="135dp"
         android:layout_marginLeft="150dp"
         android:textSize="20dp"
         android:ems="10"
         android:inputType="textPassword" />
   </RelativeLayout>

</LinearLayout>
```

Result: The program has executed successfully and output is verified

Date: 15/07/25

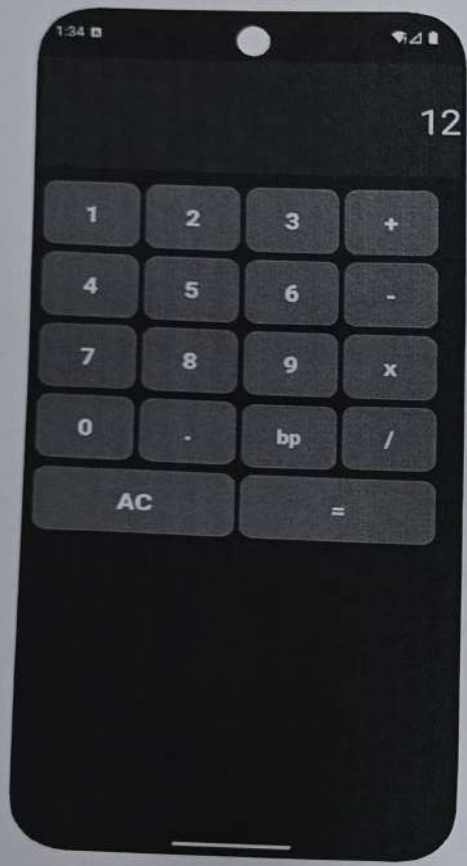
APP 2: Simple Calculator

Aim: Create a simple calculator to perform arithmetic operations

Program Code:

MainActivity.java

```
package com.example.calculator;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
public class MainActivity extends AppCompatActivity
{
  EditText t1;
  Button bac;
  String firstNumber="",operator="";
  @Override
  protected void onCreate(Bundle savedInstanceState)
  {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    t1=(EditText) findViewById(R.id.t1);
    bac=(Button) findViewById(R.id.bac);
    int b[]=new int[17];
    for(int i=0;i<=16;i++)
    {
      String s="b"+(i);
      int rid=getResources().getIdentifier(s,"id",getPackageName());
      b[i]=rid;
    }
    for(int id:b)
    {
      Button button=findViewById(id);
      if(button !=null)
      {
        button.setOnClickListener(new View.OnClickListener()
        {
          @Override
          public void onClick(View view)
```




```
        {

            handleButtonClick(view.getId());
        }});
    }}

    bac.setOnClickListener(new View.OnClickListener()
    {
      @Override
      public void onClick(View view)
      {
        t1.setText("");
      }});
}

private void handleButtonClick(int id)
{
  String txt = t1.getText().toString();
  if (id == R.id.b16)
  {
    if (!firstNumber.isEmpty() && !operator.isEmpty())
    {
      String secondNumber = txt;
      double result = 0;
      double num1 = Double.parseDouble(firstNumber);
      double num2 = Double.parseDouble(secondNumber);

      switch (operator)
      {
        case "+":
          result = num1 + num2;
          break;
        case "-":
          result = num1 - num2;
          break;
        case "x":
          result = num1 * num2;
          break;
        case "/":
          result = num1 / num2;
          break;
      }
```

```
            t1.setText(String.valueOf(result));
            firstNumber = "";
            operator = "";
        }}
      else if (id == R.id.b10)
        if (!txt.contains("."))
t1.setText(txt + ((Button) findViewById(id)).getText());
      else if (id == R.id.b12 || id == R.id.b13 || id == R.id.b14 || id == R.id.b15)
        if (!txt.isEmpty())
        {
          firstNumber = txt;
          operator = ((Button) findViewById(id)).getText().toString();
          t1.setText("");
        }
      else if (id == R.id.b11)
        if (txt.length() > 0)
          t1.setText(txt.substring(0, txt.length() - 1));
      else
        t1.setText(txt + ((Button) findViewById(id)).getText());
  }}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools
"  android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:background="#000000"
   tools:context=".MainActivity">

   <GridLayout
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:columnCount="4"
      android:rowCount="6"
      android:layout_margin="5dp">

      <EditText
         android:id="@+id/t1"
         android:layout_marginTop="350dp"
         android:layout_marginBottom="15dp"
         android:layout_width="match_parent"
         android:layout_height="150dp"
         android:layout_row="0"
         android:layout_column="0"
         android:ems="10"
         android:layout_columnSpan="4"
         android:inputType="text"
         android:textSize="92px"
         android:textAlignment="viewEnd"
         android:background="#20ffffff"
         android:textColor="#ffffff"
         android:text=""/>
      android:background="@drawable/button_background"
         android:gravity="center"
         android:text="2"
         android:textAlignment="center"
         android:textSize="22dp"
         android:textStyle="bold"
```

```
        android:layout_marginBottom="10dp"/>

    <Button
        android:id="@+id/b3"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_row="1"
        android:layout_column="2"
        android:layout_marginLeft="5dp"
android:background="@drawable/button_back
ground"
        android:text="3"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
       android:layout_marginBottom="10dp"/>

    <Button
        android:id="@+id/b12"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="1"
        android:layout_column="3"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
      android:background="@drawable/op_bg"
        android:textColor="#ffffff"
        android:text="+"/>

    <Button
        android:id="@+id/b4"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="2"
        android:layout_column="0"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="4"/>

    <Button
        android:id="@+id/b5"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="2"
        android:layout_column="1"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="5"/>

    <Button
        android:id="@+id/b6"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="2"
        android:layout_column="2"
android:background="@drawable/button_back
ground" android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
        android:text="6"/>

    <Button
        android:id="@+id/b13"
        android:layout_width="105dp"
        android:layout_height="
        android:layout_marg
```

```
android:layout_row="2"
        android:layout_column="3"
android:background="@drawable/op_bg"
        android:textColor="#ffffff"
android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
        android:text="."/>

    <Button
        android:id="@+id/b7"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="3"
        android:layout_column="0"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="7"/>

    <Button
        android:id="@+id/b8"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="3"
        android:layout_column="1"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="8"/>

    <Button
        android:id="@+id/b9"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="3"
        android:layout_column="2"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="9"/>

    <Button
        android:id="@+id/b14"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="3"
        android:layout_column="3"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/op_bg"
        android:textColor="#ffffff"
        android:text="x"/>

    <Button
        android:id="@+id/b0"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="0"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="0"/>
    <Button
```

```
        android:id="@+id/b10"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="1"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
    <Button
        android:id="@+id/b10"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="1"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground"android:text="."/>

    <Button
        android:id="@+id/b11"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="2"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="bp"/>

    <Button
        android:id="@+id/b15"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="3"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"

android:background="@drawable/op_bg"
        android:textColor="#ffffff"
        android:text="/"/>
        <Button
        android:id="@+id/bac"
        android:layout_width="215dp"
        android:layout_height="75dp"
        android:layout_row="5"
        android:layout_column="0"
        android:layout_columnSpan="2"
        android:layout_marginLeft="5dp"
        android:text="AC"
        android:textColor="#ffffff"
android:background="@drawable/op_bg"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
android:layout_marginBottom="10dp"/>

    <Button
        android:id="@+id/b16"
        android:layout_width="215dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="5"
        android:layout_column="2"
        android:layout_columnSpan="2"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
```

```
android:background="@drawable/op_bg"
    android:textColor="#ffffff"
    android:text="=" />
```

```
</GridLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified

Date: 22/07/25

APP 3: Intent

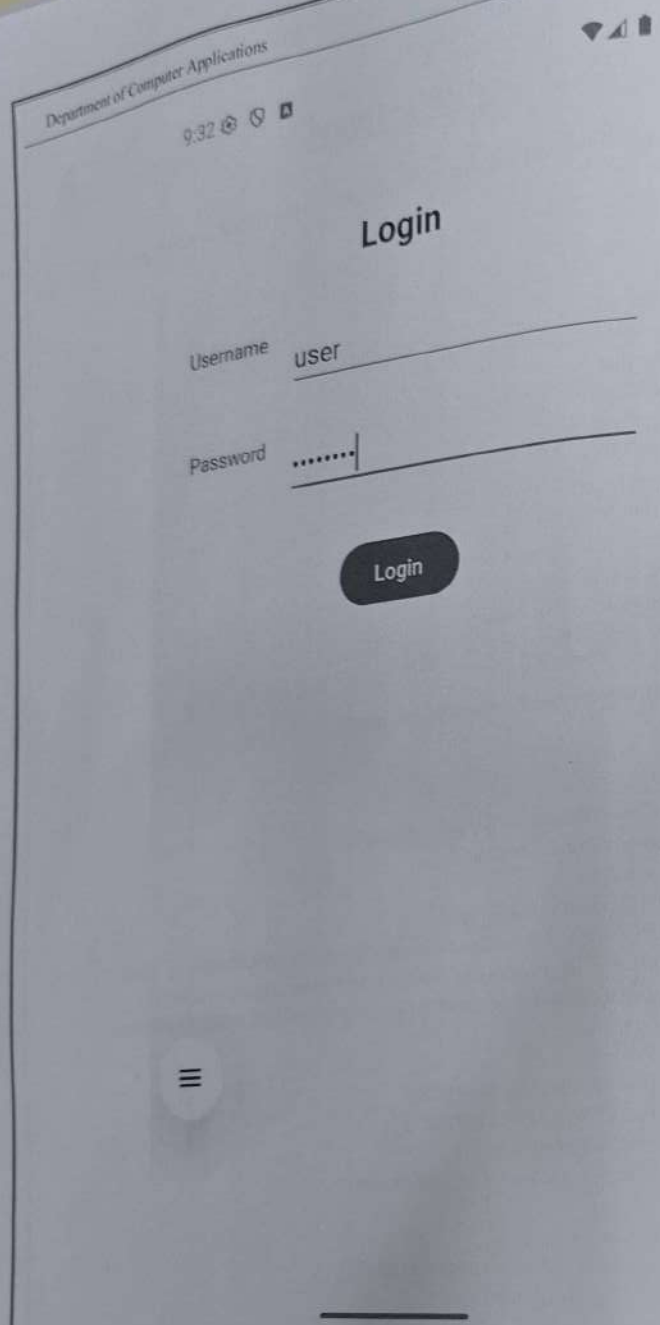
Aim: Create two activities , 1st activity contains login details, when clicked to submit, go to 2nd activity.

Program Code:

MainActivity.java

```
package com.example.login;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    EditText e1,e2;
    Button b1;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        e1=(EditText) findViewById(R.id.edittext1);
        e2=(EditText) findViewById(R.id.edittext2);
        b1=(Button) findViewById(R.id.btn1);
        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String username=e1.getText().toString().trim();
                String pass=e2.getText().toString();
```




```
        if(username.equals("user") && pass.equals("admin123")) {
            Toast.makeText(MainActivity.this, "Login successful", Toast.LENGTH_SHORT).show();
            Intent intent = new Intent(MainActivity.this, Homepage.class);
            startActivity(intent);
        }
        else
            Toast.makeText(MainActivity.this,"Login failed",Toast.LENGTH_SHORT).show();
    }
  });
 }}
```

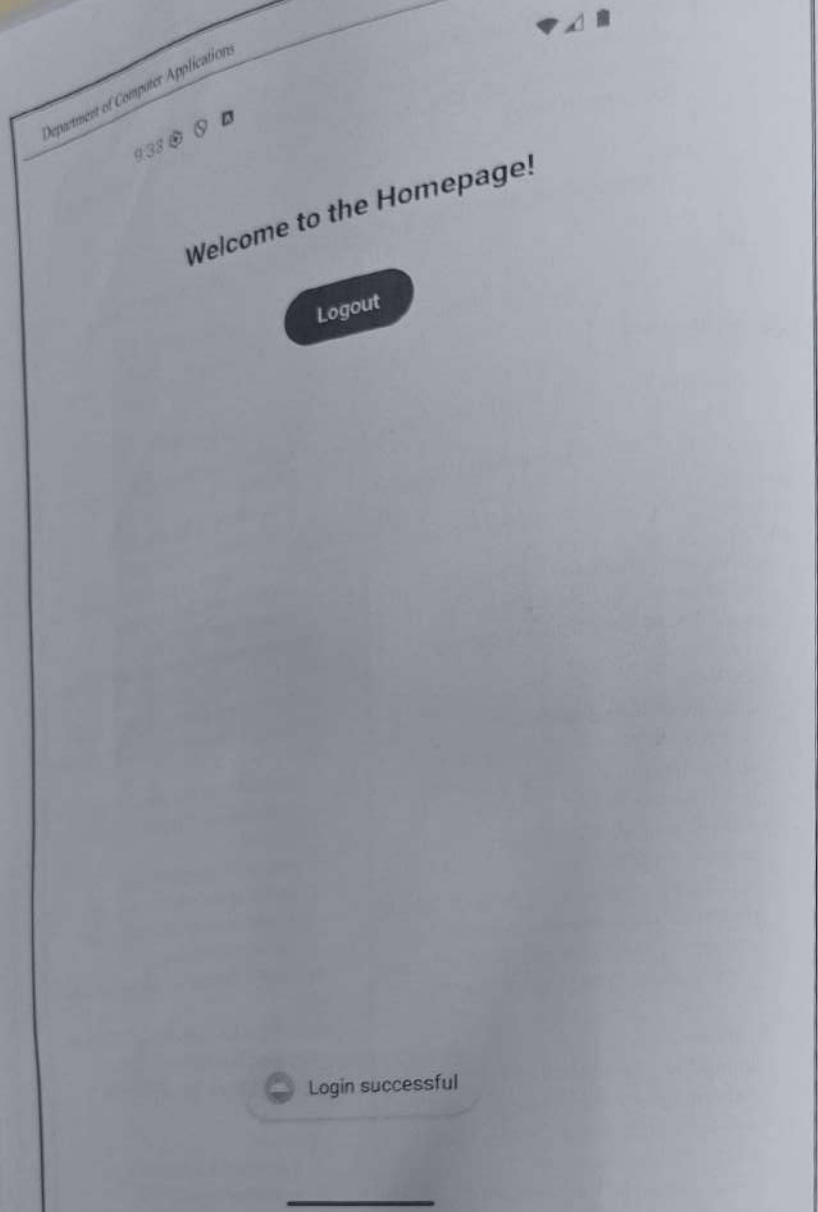
Activity_Main.xml

```
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

  <TextView
    android:id="@+id/textView3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login"
    android:textSize="24sp"
    android:textStyle="bold"
    app:layout_constraintTop_toTopOf="parent"
   app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="64dp" />

  <TextView
    android:id="@+id/textView4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Username"
app:layout_constraintTop_toBottomOf="@id/textView3"
   app:layout_constraintStart_toStartOf="parent"
    android:layout_marginTop="32dp"
    android:layout_marginStart="32dp" />
  <EditText
    android:id="@+id/edittext1"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:hint="Enter username"
    android:inputType="text"
app:layout_constraintStart_toEndOf="@id/textView4"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintTop_toTopOf="@id/textView4"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="32dp" />
  <TextView
    android:id="@+id/textView5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Password"
app:layout_constraintTop_toBottomOf="@id/edittext1"
app:layout_constraintStart_toStartOf="parent"
    android:layout_marginTop="24dp"
    android:layout_marginStart="32dp" />

  <EditText
    android:id="@+id/edittext2"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:hint="Enter password"
    android:inputType="textPassword"
```




```
app:layout_constraintStart_toEndOf="@id/textVi
ew5"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintTop_toTopOf="@id/textVie
w5"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="32dp" />

  <Button
    android:id="@+id/btn1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login"
app:layout_constraintTop_toBottomOf="@id/edit
text2"
app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="32dp" />
</androidx.constraintlayout.widget.ConstraintLay
out>
```

Homepage.java

```
package com.example.login;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Homepage extends AppCompatActivity {
  Button logoutButton;
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_homepage);
    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
      Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
      v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
      return insets;
    });

    logoutButton = findViewById(R.id.logoutButton);
    logoutButton.setOnClickListener(new View.OnClickListener() {
      @Override
    public void onClick(View v) {
        // Go back to login screen
```

```
        Intent intent = new Intent(Homepage.this, MainActivity.class);
        startActivity(intent);
        finish();
      }
    });
  }}
```

Activity_HomePage.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintL
ayout
xmlns:android="http://schemas.android.com/ap
k/res/android"
xmlns:app="http://schemas.android.com/apk/re
s-auto"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:id="@+id/main"
  android:padding="24dp">

  <TextView
    android:id="@+id/welcomeText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Welcome to the
Homepage!"
    android:textSize="20sp"
    android:textStyle="bold"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="64dp" />
  <Button
    android:id="@+id/logoutButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Logout"
app:layout_constraintTop_toBottomOf="@id/
welcomeText"

app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="32dp" />
</androidx.constraintlayout.widget.Constraint
Layout>
```

Result: The program has executed successfully and output is verified

Date : 05/08/25

APP 4: Activity Life Cycle

Aim: Create an app to demonstrate Activity Life Cycle.

Program Code :

MainActivity.java

```
package com.example.start;
import androidx.appcompat.app.AppCompatActivity;
import android.nfc.Tag;
import android.os.Bundle;
import android.util.Log;
public class MainActivity extends
AppCompatActivity
{
@Override
protected void onCreate(Bundle savedInstanceState)
{
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
Log.d("Appstate","Created");
}
@Override
protected void onStart()
{
super.onStart();
Log.d("Appstate","Starting");
}
@Override
protected void onResume()
{
super.onResume();
Log.d("Appstate","Resuming");
}
@Override
protected void onPause()
{
super.onPause();
Log.d("Appstate","Paused");
}
@Override
protected void onStop()
{
super.onStop();
Log.d("Appstate","Stopped");
}
@Override
protected void onRestart()
{
super.onRestart();
Log.d("Appstate","Restarted");
}
@Override
protected void onDestroy()
{
super.onDestroy();
Log.d("Appstate","Destroyed");
}
}
```

Result: The program has executed successfully and output is verified

Date : 12/08/25

APP 5: Checkbox and Radiobutton

Aim : Create an app to demonstrate the usage of Checkbox and Radio buttons

Program Code :

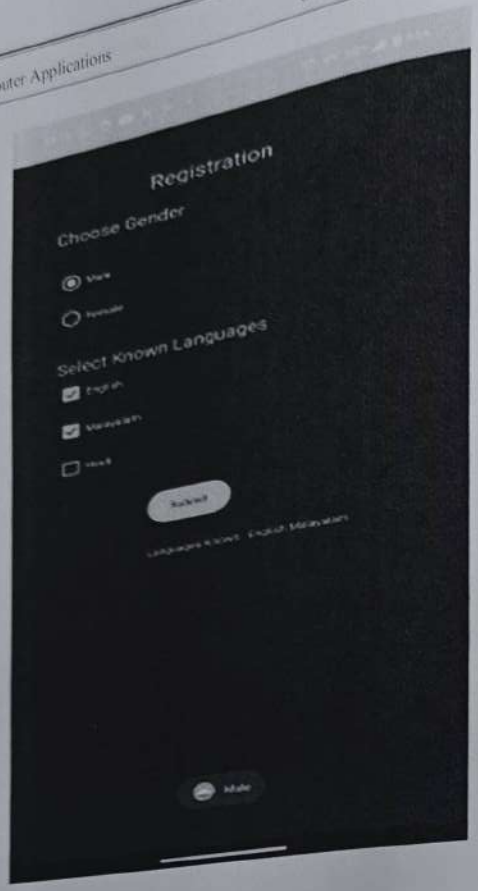
MainActivity.java:

```
package com.example.radiocheck;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity
{
Button b1;
RadioGroup rg;
RadioButton r1,r2;
CheckBox e,m,h;
TextView t;
@Override
protected void onCreate(Bundle savedInstanceState)
{
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
b1=(Button) findViewById(R.id.submit);
r1=(RadioButton) findViewById(R.id.male);
r2=(RadioButton) findViewById(R.id.female);
e=(CheckBox)findViewById(R.id.e);
m=(CheckBox)findViewById(R.id.m);
h=(CheckBox)findViewById(R.id.h);
t=(TextView)findViewById(R.id.tv);
b1.setOnClickListener(new View.OnClickListener()
{
@Override
public void onClick(View view)
{
String s[]=new String[3];
t.setText("Languages Known: ");
if(e.isChecked())
t.setText(t.getText().toString()+" "+e.getText());
if(m.isChecked())
t.setText(t.getText().toString()+" "+m.getText());
```

```
if(h.isChecked())
t.setText(t.getText().toString()+" "+h.getText());
}
});
rg=(RadioGroup) findViewById(R.id.rg);
rg.setOnCheckedChangeListener(new RadioGroup.OnCheckedChangeListener()
{
@Override
public void onCheckedChanged(RadioGroup group, int checkedId)
{
if(checkedId==R.id.female)
{ Toast.makeText(getApplicationContext(),r2.getText().toString(),Toast.LENGTH_SHORT).show();
}
else
{ Toast.makeText(getApplicationContext(),r1.getText().toString(),Toast.LENGTH_SHORT).show();
}
}
});
}
}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tool"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<RelativeLayout
android:layout_width="match_parent"
android:layout_height="match_parent">
<TextView
android:layout_width="wrap_content"
android:layout_height="30dp"
android:layout_marginTop="50dp"
android:layout_marginLeft="150dp"
android:id="@+id/h1"
android:text="Registration"
android:textSize="24dp"/>
<TextView
android:layout_width="wrap_content"
android:layout_height="30dp"
android:layout_marginTop="100dp"
android:layout_marginLeft="50dp"
android:id="@+id/gender"
android:text="Choose Gender"
android:textSize="20dp"/>
<RadioGroup
android:layout_width="250dp"
android:layout_height="wrap_content"
android:id="@+id/rg"
android:layout_marginTop="150dp"
android:layout_marginLeft="50dp">
<RadioButton
android:layout_width=
"wrap_content"
android:layout_height=
"wrap_content"
android:text="Male"
android:id="@+id/male"/>
<RadioButton
android:layout_width=
"wrap_content"
android:layout_height=
"wrap_content"
```




```
android:text="Female"
android:id="@+id/female"/>
</RadioGroup>
<TextView
android:layout_width="wrap_content"
android:layout_height="30dp"
android:layout_marginTop="275dp"
android:layout_marginLeft="50dp"
android:id="@+id/lang"
android:text="Select Known Languages"
android:textSize="20dp"/>
<CheckBox
android:id="@+id/e"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="English"
android:layout_marginTop="300dp"
android:layout_marginLeft="50dp"/>
<CheckBox
android:id="@+id/m"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Malayalam"
android:layout_marginTop="350dp"
android:layout_marginLeft="50dp"/>
<CheckBox
android:id="@+id/h"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Hindi"
android:layout_marginTop="400dp"
android:layout_marginLeft="50dp"/>
<Button
android:layout_width="wrap_content"
android:layout_height="50dp"
android:layout_marginTop="475dp"
android:layout_marginLeft="150dp"
android:text="Submit"
android:id="@+id/submit"/>
```

```
<TextView
android:layout_width="wrap_content"
android:layout_height="50dp"
android:layout_marginTop="550dp"
android:layout_marginLeft="150dp"
android:text=""
android:id="@+id/tv"/>
</RelativeLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified.

Date : 19/08/25

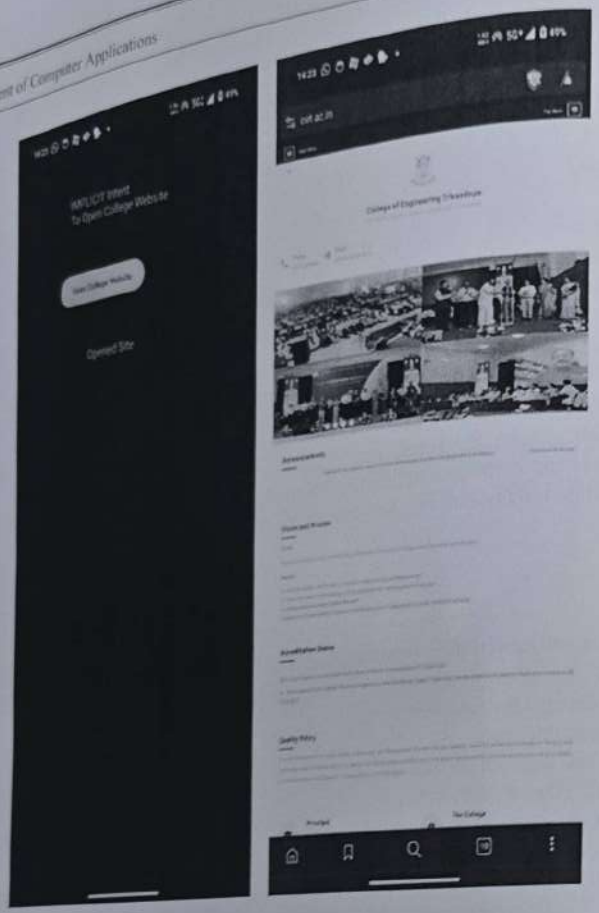
APP 6: Implicit Intent

Aim : Create an app to demonstrate the usage of implicit intent by opening a website form the app

Program Code :

MainActivity.java:

```
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity
{
Button b;
TextView t;
@Override
protected void onCreate(Bundle savedInstanceState)
{
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);
b=(Button) findViewById(R.id.b1);
t=(TextView) findViewById(R.id.t1);
b.setOnClickListener(new View.OnClickListener()
{
@Override
public void onClick(View view) {
Intent intent = new Intent(Intent.ACTION_VIEW);
intent.setData(Uri.parse("https://www.cet.ac.in"));
startActivity(intent);
t.setText("Youtube Opened");
}
});
}
}
```



Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools" android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

<RelativeLayout
android:layout_width="wrap_content"
android:layout_height="wrap_content">

<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="IMPLICIT Intent \nTo Open College Website"
android:layout_marginLeft="115dp"
android:layout_marginTop="100dp"
android:textSize="20sp">
</TextView>

<Button
android:id="@+id/b1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="95dp"
android:layout_marginTop="200dp"
android:text="Open College Website">
</Button>

<TextView
android:id="@+id/t1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text=""
android:layout_marginTop="300dp"
android:layout_marginLeft="150dp"
android:textSize="20sp"></TextView>
</RelativeLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified

Date : 26/08/25

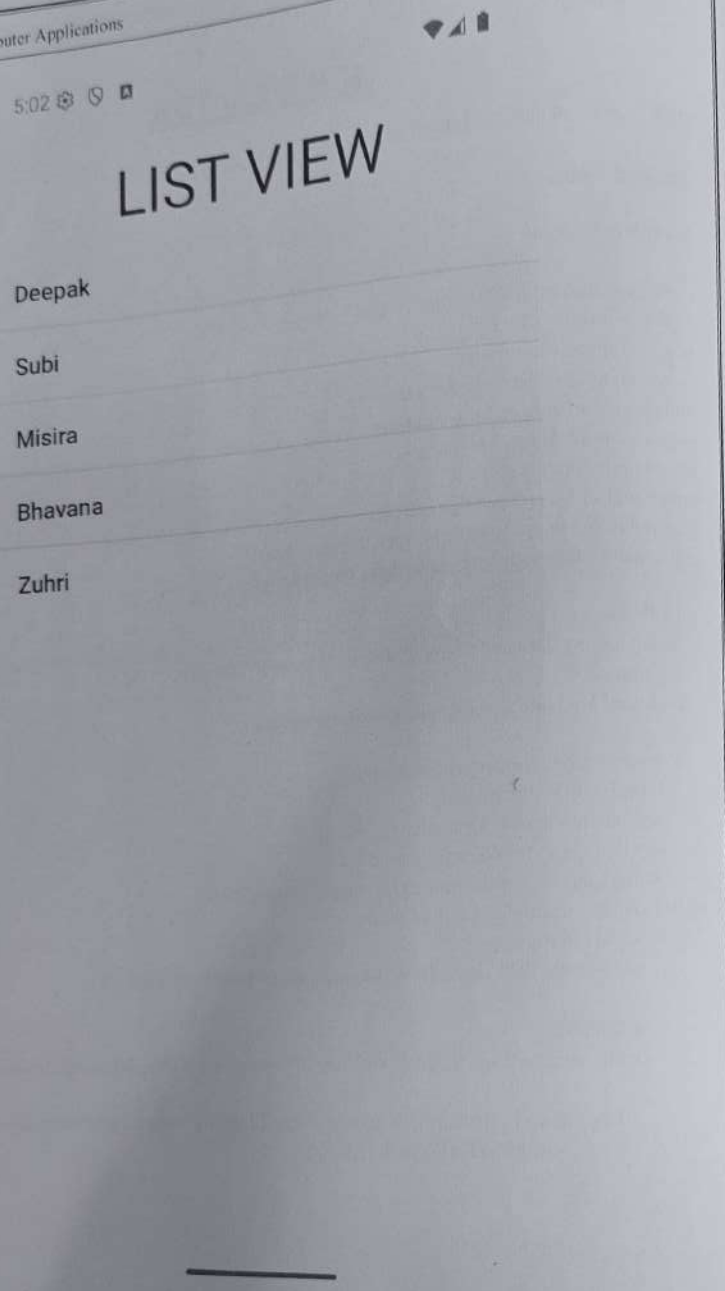
APP 7: List View

Aim : Create an app to create a List view with few items in it

Program Code :

MainActivity.java:

```
package com.example.listview;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity
{
   ListView lv;
   String arr[]={"Deepak","Subi","Misira","Bhavana","Zuhri"};
   @Override
   protected void onCreate(Bundle savedInstanceState)
   {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);
      lv=(ListView) findViewById(R.id.lv);
      ArrayAdapter adapter=new ArrayAdapter<String>(this,
android.R.layout.simple_list_item_1,arr);
      lv.setAdapter(adapter);
      lv.setOnItemClickListener(new AdapterView.OnItemClickListener()
      {
         @Override
         public void onItemClick(AdapterView<?> parent, View view, int position, long id)
         {
            Toast.makeText(MainActivity.this,"Name Clicked: "+adapter.getItem(position),
                  Toast.LENGTH_SHORT).show();
         }
      });
   }
}
```



Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
 android:id="@+id/main"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity">

<RelativeLayout
 android:layout_width="wrap_content"
 android:layout_height="wrap_content">

 <TextView
android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_marginTop="50dp"
 android:layout_marginLeft="100dp"
 android:text="LIST VIEW"
 android:textSize="42dp">
</TextView>

 <ListView
 android:id="@+id/lv"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_marginTop="115dp"
 android:layout_marginBottom="50dp"
 android:layout_marginLeft="10dp">
</ListView>

</RelativeLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified.

Date : 09/09/25

APP 8 : Scroll View

Aim : Create an app to create a spinner and select an item from it.

Program Code :

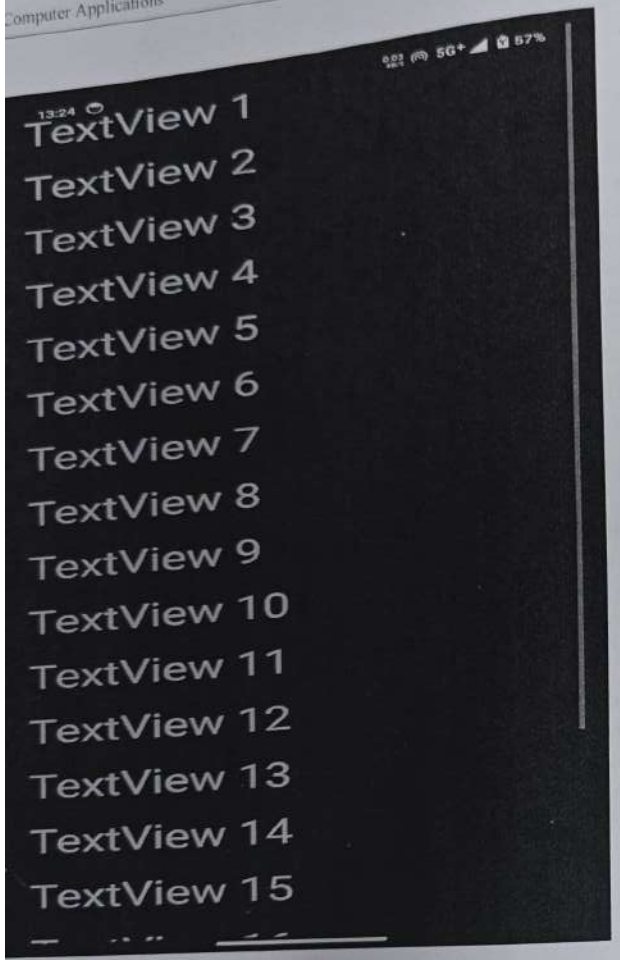
MainActivity.java:

```
package com.example.scrollview;
import android.os.Bundle;
import android.widget.LinearLayout;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity
{
@Override
protected void onCreate(Bundle savedInstanceState)
{
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);
LinearLayout linearLayout = findViewById(R.id.linearLayoutContainer);
for (int i = 1; i <= 20; i++)
{
TextView textView = new TextView(this);
textView.setText("TextView " + i);
textView.setTextSize(45);
textView.setPadding(16, 16, 16, 16);
linearLayout.addView(textView);
}}}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
 android:id="@+id/main"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 android:padding="16dp"
 tools:context=".MainActivity">
 <LinearLayout
 android:id="@+id/linearLayoutContainer"
 android:layout_width="match_parent"
 android:layout_height="wrap_content"
 android:orientation="vertical">
 </LinearLayout>
</ScrollView>
```

Result: The program has executed successfully and output is verified



Date : 16/09/25

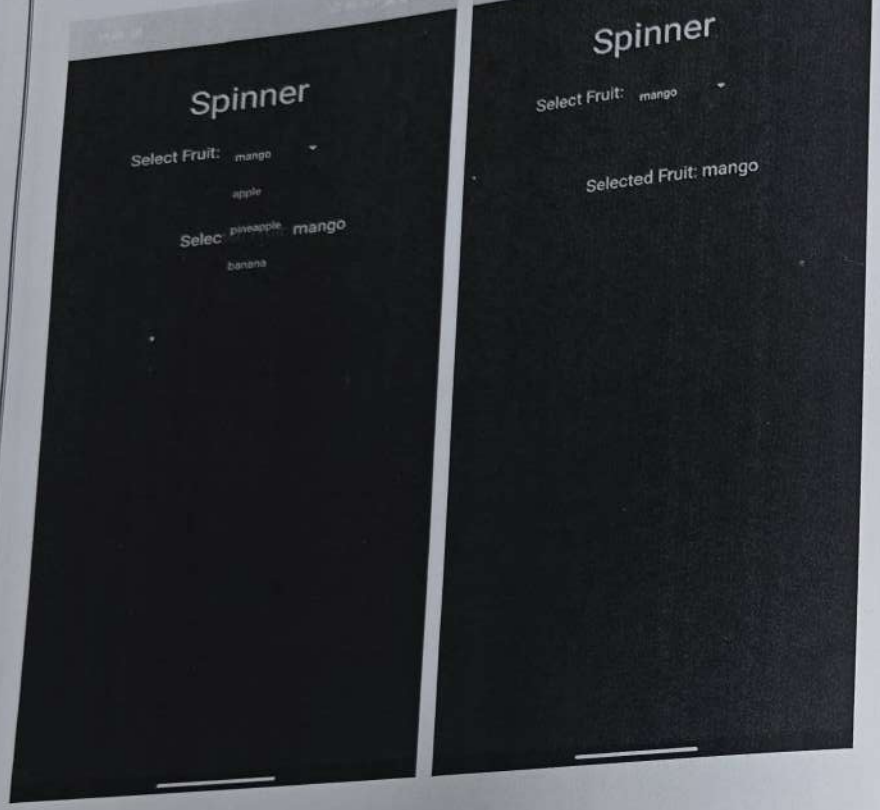
APP 9 : Spinner

Aim : Create an app to create a spinner and select an item from it.

Program Code :

MainActivity.java:

```
package com.example.spinner;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
import android.widget.TextView;
import android.widget.Toast;
import java.util.ArrayList;
public class MainActivity extends AppCompatActivity
{
Spinner s;
TextView t1;
@Override
protected void onCreate(Bundle savedInstanceState)
{
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
s=(Spinner)findViewById(R.id.sp);
t1=(TextView) findViewById(R.id.t1);
ArrayList <String> al=new ArrayList<String>();
al.add("mango");
al.add("apple");
al.add("pineapple");
al.add("banana");
ArrayAdapter ad=new ArrayAdapter<>(this, androidx.appcompat.R.layout.support_simple_
spinner_dropdown_item,al);
s.setAdapter(ad);
s.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener()
{
@Override
public void onItemSelected(AdapterView<?> parent, View view, int position, long id)
{
t1.setText("Selected Fruit: "+al.get(position));
}
```




```
@Override
public void onNothingSelected(AdapterView<?> parent)
{
t1.setText("Select a Fruit !!!!");
}
});
}
}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<RelativeLayout
android:layout_width="wrap_content"
android:layout_height="wrap_content">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="50dp"
android:text="Spinner"
android:textSize="42dp"></TextView>
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="85dp"
android:layout_marginTop="135dp"
android:text="Select Fruit:"
android:textSize="20dp">
</TextView>
<Spinner
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="200dp"
android:layout_marginTop="135dp"
android:id="@+id/sp"></Spinner>
<TextView
android:id="@+id/t1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="250dp"
android:text=""
android:textSize="20dp"></TextView>
</RelativeLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified.

Date : 23/09/25

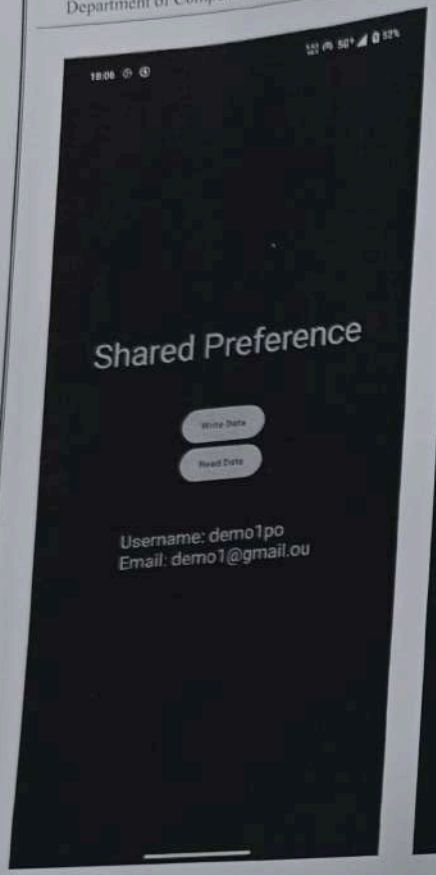
APP 10 : Shared Preferances

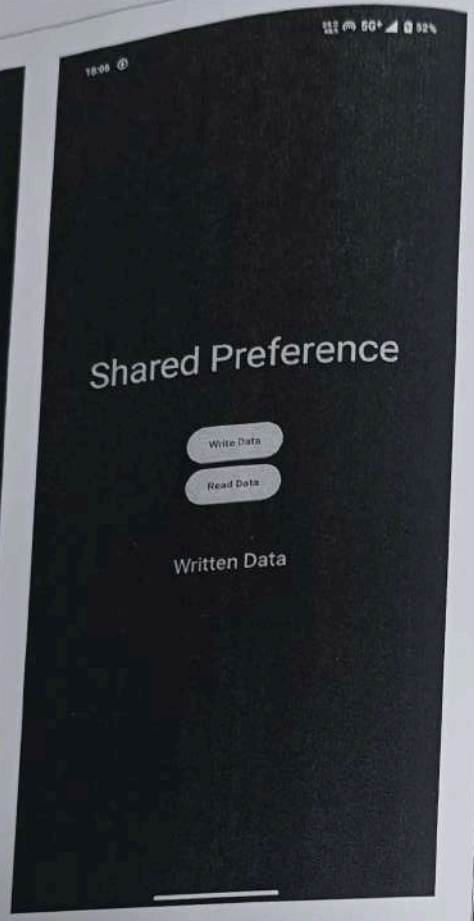
Aim : Create an app to demonstrate the usage of shared preferences by writing and reading data.

Program Code :

MainActivity.java

```
import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity
{
TextView txt;
Button br,bw;
@Override
protected void onCreate(Bundle savedInstanceState)
{
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);
txt=(TextView) findViewByld(R.id.text_status);
bw=(Button) findViewById(R.id.btn_write);
br=(Button) findViewById(R.id.btn_read);
String usernamedemo="demo1po";
String emaildemo="demo1@gmail.ou";
SharedPreferences sp =getSharedPreferences("sp_abc",MODE_PRIVATE);
bw.setOnClickListener(new View.OnClickListener()
{
@Override
public void onClick(View view)
{
SharedPreferences.Editor se = sp.edit();
se.putString("username",usernamedemo);
se.putString("email",emaildemo);
se.apply();
txt.setText("Written Data");
}
});
```






```
br.setOnClickListener(new View.OnClickListener()
{
@Override
public void onClick(View view) {
String u,e;
u=sp.getString("username","");
e=sp.getString("email","");
txt.setText("Username: "+u+" \nEmail: "+e);
}
});
}
}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools
"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity"
android:orientation="vertical"
android:gravity="center">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Shared Preference"
android:textSize="42dp"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Write Data"
android:layout_marginTop="50dp"
android:id="@+id/btn_write"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Read Data"
android:id="@+id/btn_read"/>
<TextView
android:id="@+id/text_status"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="50dp"
android:text="STATUS"
android:textSize="22dp"/>

</LinearLayout>
```

Result: The program has executed successfully and output is verified.